LHCb Open Data Analysis

A CERN Open Data Analysis of Three-Body B-Meson Decays

Invariant-Mass Reconstruction, Signal Classification, CP-Asymmetry Diagnostics, and Dalitz-Plot Studies

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Portfolio report for experimental particle-physics PhD applications

Abstract

This report presents a compact but realistic high-energy-physics analysis workflow based on LHCb CERN Open Data. Three-body B-meson candidates are reconstructed from charged-hadron tracks under a kaon mass hypothesis. The workflow uses ROOT input files, Python-based columnar data processing, invariant-mass reconstruction, event selection, a sideband-labelled Random Forest classifier, a raw charge-asymmetry diagnostic, and Dalitz-plot visualisations. The report emphasizes reproducibility, modular scientific code, and analysis reasoning suitable for an early-stage experimental particle-physics project.

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Contents

1 Introduction **1**

2 The LHCb Experiment **1**

3 Dataset and Software Framework **2**
3.1 ROOT input files 2
3.2 Python analysis workflow 2

4 Invariant Mass Reconstruction **3**
4.1 Relativistic mass formula 3
4.2 Mass spectrum before and after selection 3
4.3 Signal-plus-background fit 4

5 Event Selection **5**

6 Signal and Background Classification **6**
6.1 Sideband-labelled samples 6
6.2 Random Forest classifier 6

7 CP Asymmetry Study **8**
7.1 Charge separation 8
7.2 Raw integrated asymmetry 9

8 Dalitz Plot Analysis **9**
8.1 Pairwise invariant masses 9
8.2 Dalitz distributions and resonance structure 10

9 Conclusion **12**

1 Introduction

The Standard Model of particle physics describes the known elementary particles and their interactions with remarkable precision. Nevertheless, several observed features of the Universe remain only partially understood. One of the most important is the matter–antimatter asymmetry: cosmological observations indicate that the visible Universe is dominated by matter, while the known laws of particle physics contain only limited sources of charge-parity (CP) violation. Precision studies of flavour-changing processes are therefore a central part of modern particle physics.

Flavour physics studies transitions between different quark flavours. In the heavy-flavour sector, hadrons containing beauty quarks provide especially rich laboratories. The large mass of the beauty quark allows many decay modes, including multibody final states in which resonant and non-resonant amplitudes can interfere. Such interference can generate local CP-asymmetric effects in restricted regions of phase space, even when the integrated asymmetry is small.

B mesons are bound states containing a beauty quark or antiquark and a lighter valence quark. Charged B mesons, denoted B^+ and B^-, form a natural pair of charge-conjugate systems. By comparing the observed yields or distributions of B^+ and B^- candidates, one can construct raw asymmetry observables. In a simplified counting analysis, the integrated asymmetry is written as

$$A_{\mathrm{CP}}^{\mathrm{raw}} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}, \tag{1}$$

where $N(B^+)$ and $N(B^-)$ are observed candidate counts. A full physics measurement must correct this raw quantity for background contributions, detector-induced charge asymmetries, production asymmetries, trigger effects, and selection efficiencies. The present project treats Eq. 1 as an analysis diagnostic rather than as a final CP-violation measurement.

This report documents a reproducible analysis workflow built with CERN Open Data [1]. The project focuses on a sample of reconstructed LHCb three-hadron B candidates stored in ROOT files. The analysis reconstructs the B-candidate invariant mass, applies simple physics-motivated selections, fits the mass peak, trains a signal/background classifier using sideband labels, separates charge-conjugate candidates, and produces Dalitz-plot distributions. The goal is to demonstrate a modern high-energy-physics (HEP) workflow using transparent scientific reasoning and modular Python code.

The analysis is intentionally compact. It is not presented as a publishable LHCb measurement, but as a polished student-level project that mirrors the structure of a real experimental analysis: data access, reconstruction, selection, modelling, validation plots, and interpretation of limitations. This makes it suitable as a technical portfolio project for applications in experimental particle physics.

2 The LHCb Experiment

The Large Hadron Collider beauty experiment (LHCb) is a forward spectrometer at the CERN Large Hadron Collider designed for precision studies of hadrons containing beauty and charm quarks [2]. Proton–proton collisions at the LHC produce heavy quarks predominantly in the forward and backward regions. LHCb exploits this production topology by covering a forward angular region, providing excellent tracking, vertexing, momentum resolution, and particle identification.

The detector design is particularly well suited to heavy-flavour analyses. B hadrons have lifetimes long enough to travel measurable distances before decaying. This displaced-decay topology is identified by reconstructing primary vertices from the proton–proton interaction and secondary vertices from the B-hadron decay products. The distance between these vertices and the quality of the reconstructed secondary vertex are powerful handles for suppressing prompt combinatorial background.

Charged-particle tracking is performed with a system of silicon and tracking stations around a dipole magnet. The curvature of charged-particle trajectories in the magnetic field provides a momentum measurement and determines the charge sign. In the present analysis, the track momentum components are used directly to build four-vectors for the three daughter hadrons. The charge information is used later to classify candidates as B^+ or B^-.

Particle identification (PID) is another central ingredient. LHCb uses ring-imaging Cherenkov detectors to distinguish kaons, pions, and protons over a broad momentum range. The open-data ntuples used in this project contain per-track PID-related variables such as kaon and pion probability-like observables. A simple kaon-PID selection is applied to build a clean demonstration sample under a $B \to KKK$ mass hypothesis.

In a complete LHCb analysis, detector effects are modelled with simulation, calibration samples, and data-driven corrections. This project keeps the detector treatment deliberately simple, but it preserves the main conceptual ingredients: displaced B vertices, charged-track reconstruction, particle identification, and magnet-polarity awareness.

3 Dataset and Software Framework

3.1 ROOT input files

The project uses CERN Open Data ROOT files containing reconstructed B-candidate ntuples. The main file used in the notebooks is

`data/raw/B2HHH_MagnetDown.root`.

Additional files for future extensions are stored alongside it:

`B2HHH_MagnetUp.root`, `PhaseSpaceSimulation.root`.

The presence of both magnet polarities is important for realistic charge-asymmetry studies, because reversing the magnetic field helps control detector-induced left–right and charge-dependent reconstruction effects. The present report focuses on the MagnetDown sample for a compact demonstration.

The data are stored in the ROOT format [3], a standard file and object format in high-energy physics. The relevant event information is contained in a tree named `DecayTree`. A ROOT `TTree` stores columnar data: each branch corresponds to one observable, such as a momentum component, charge, PID response, vertex quality, or flight distance. This format is efficient for large event samples because only the branches needed for a given analysis step need to be read.

3.2 Python analysis workflow

The analysis is implemented as a Python workflow, with reusable functions under `src/` and exploratory or result-producing notebooks under `notebooks/`. The main software components are listed in Table 1.

Table 1: Main software tools used in the analysis workflow.

Tool	Role in the analysis
ROOT	Standard HEP file format and ntuple storage
`uproot`	Reading ROOT TTrees directly in Python
`pandas`	Tabular event manipulation and selections
`numpy`	Vectorised numerical calculations
`matplotlib`	Publication-style figures
`scikit-learn`	Random Forest classifier and performance metrics

The branch set used throughout the analysis includes the momentum components `H1_PX`, `H1_PY`, `H1_PZ` through `H3_PZ`, the three track charges, PID variables such as `H1_ProbK`, and B-candidate quality variables such as `B_VertexChi2` and `B_FlightDistance`. A shared selection helper loads these branches, reconstructs the B-candidate mass, and applies the common baseline cuts. This avoids duplicated logic between the mass, machine-learning, CP-asymmetry, and Dalitz notebooks.

The workflow is intentionally modular. The main reusable components are the invariant-mass reconstruction module `src/reconstruction/invariant_mass.py`, the mass-plotting helpers in `src/plotting/mass_plots.py`, the signal/background utilities in `src/ml/signal_background.py`, and the Dalitz-kinematics module `src/physics/dalitz.py`. This separation mirrors the structure of a maintainable HEP analysis, where physics definitions, plotting conventions, and machine-learning utilities should be reusable across notebooks.

4 Invariant Mass Reconstruction

4.1 Relativistic mass formula

The key reconstructed observable is the invariant mass of the three-hadron B candidate. For a system of particles with total energy E and total momentum $\vec{p}$, the invariant mass is defined by

$$m^2c^4 = E^2 - |\vec{p}|^2c^2. \tag{2}$$

In high-energy-physics units with $c = 1$, this becomes

$$m^2 = E^2 - p_x^2 - p_y^2 - p_z^2. \tag{3}$$

For each daughter track, the analysis uses a charged-kaon mass hypothesis,

$$m_K = 493.677\,\mathrm{MeV}/c^2, \tag{4}$$

and reconstructs the energy as

$$E_i = \sqrt{p_{x,i}^2 + p_{y,i}^2 + p_{z,i}^2 + m_K^2}. \tag{5}$$

The B-candidate four-vector is then the sum of the three daughter four-vectors:

$$E_B = \sum_{i=1}^{3} E_i, \qquad \vec{p}_B = \sum_{i=1}^{3} \vec{p}_i. \tag{6}$$

Substituting these quantities into Eq. 3 gives the reconstructed B-candidate mass. The calculation is performed event-by-event with vectorised `numpy` operations.

4.2 Mass spectrum before and after selection

Figure 1 shows the reconstructed three-kaon mass spectrum before the baseline event selection. The distribution contains a broad combinatorial component as well as a visible accumulation near the expected B-meson mass region. This broad structure is typical of a preselection-level hadron-collider sample, where many random track combinations can satisfy a loose candidate definition.

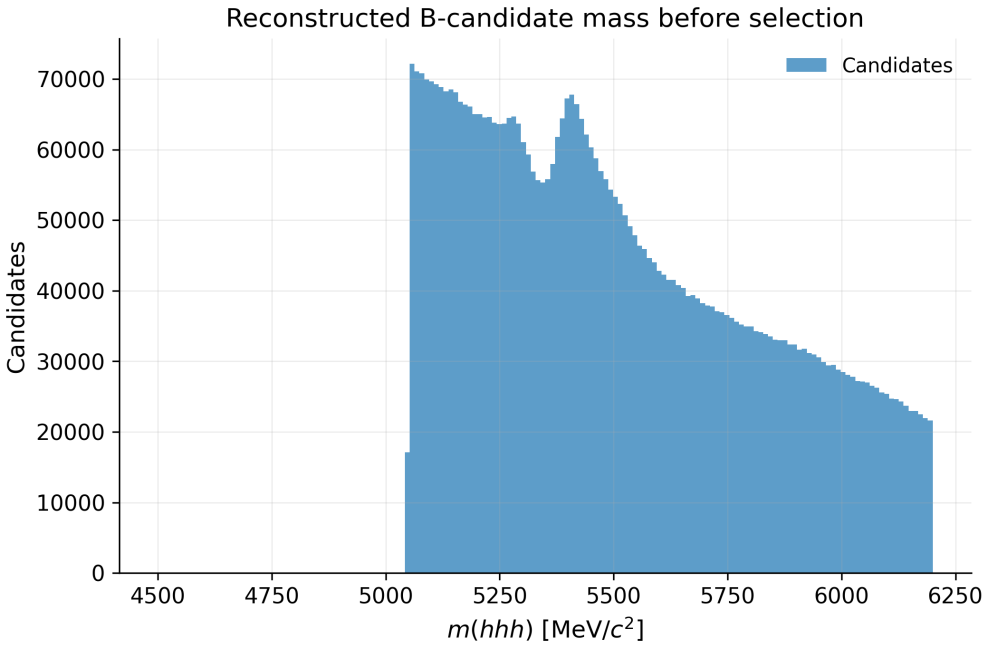


Figure 1: Reconstructed B-candidate invariant-mass distribution before baseline selection.

After applying quality, displacement, and kaon-PID selections, the mass spectrum becomes visibly cleaner, as shown in Fig. 2. The selection enhances the relative contribution of candidates compatible with a displaced B decay and suppresses a large fraction of combinatorial background.

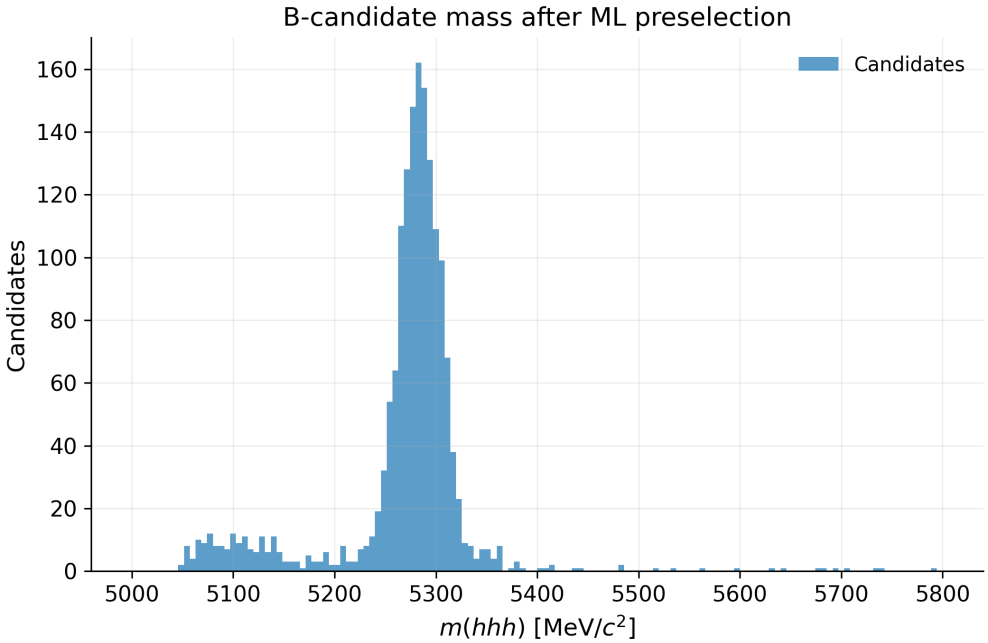


Figure 2: Reconstructed B-candidate invariant-mass distribution after baseline physics selections.

4.3 Signal-plus-background fit

A first signal extraction is performed with a simple Gaussian signal model plus a linear background model:

$$f(m) = A \exp\left[-\frac{(m-\mu)^2}{2\sigma^2}\right] + am + b. \tag{7}$$

This model is not intended to replace a full analysis fit, but it provides a useful validation of the reconstructed peak position and width. The fitted distribution is shown in Fig. 3. The visible peak near the B-meson mass demonstrates that the four-vector reconstruction and event selection are behaving sensibly.

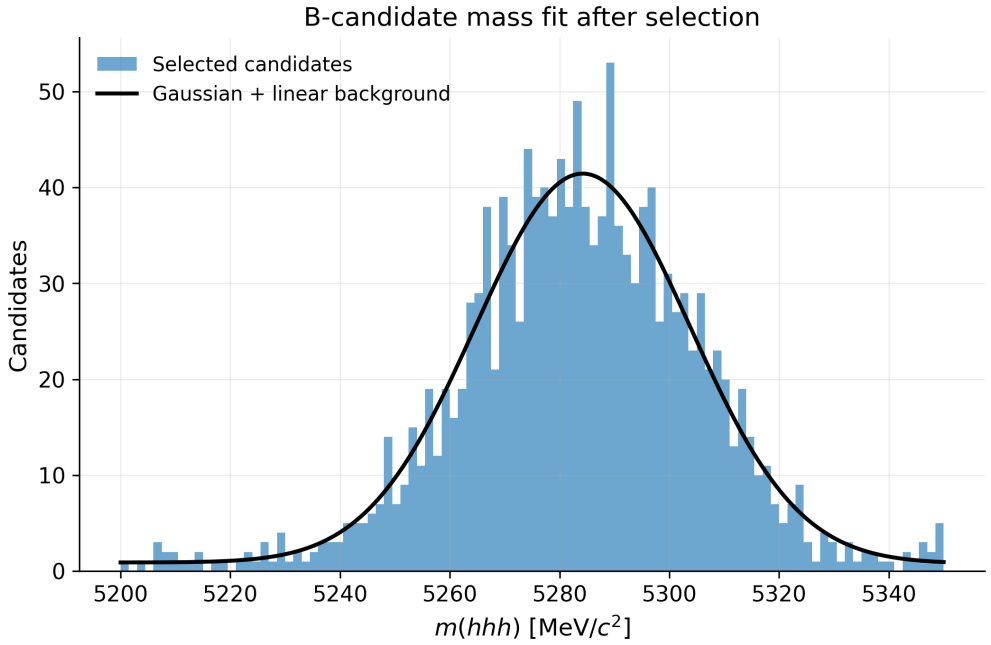


Figure 3: Selected B-candidate mass spectrum fitted with a Gaussian signal component and linear background component.

5 Event Selection

Event selection is a central part of any hadron-collider analysis. The raw candidate sample contains true signal decays, partially reconstructed backgrounds, misidentified backgrounds, and random combinations of charged tracks. The goal of the baseline selection is to retain a useful signal-rich region while keeping the cuts simple enough to interpret.

The baseline selection used in this project is

$$\chi^2_{\text{vertex}}(B) < 5, \tag{8}$$

$$d_{\text{flight}}(B) > 20, \tag{9}$$

$$P_K(H_1), P_K(H_2), P_K(H_3) > 0.5. \tag{10}$$

Here $\chi^2_{\text{vertex}}(B)$ denotes the B-candidate vertex-quality variable, $d_{\text{flight}}(B)$ denotes the reconstructed flight-distance variable, and $P_K(H_i)$ denotes the kaon PID response for daughter track H_i.

The vertex-quality cut suppresses candidates whose three tracks do not form a well-reconstructed secondary vertex. This is physically motivated because the daughters of a true B decay should originate from a common decay point. A poor vertex fit is more likely to arise from accidental combinations of tracks produced in different processes or from poorly reconstructed tracks.

The flight-distance selection exploits the finite lifetime of B mesons. Because B hadrons travel a measurable distance before decaying, their decay vertices are displaced from the primary proton–proton interaction point. Requiring a sufficiently large flight distance therefore suppresses prompt background combinations that originate near the primary vertex.

The PID selection implements the kaon mass hypothesis used in the invariant-mass reconstruction. Requiring all three tracks to have kaon-compatible PID values improves the consistency of

the reconstructed final state. In a more complete analysis, one would optimize PID requirements with calibration samples and evaluate the corresponding efficiency and misidentification rates. Here the selection is chosen as a transparent demonstration of how detector-level PID information enters the analysis.

The selected sample is used consistently across the mass reconstruction, signal/background classification, CP-asymmetry diagnostic, and Dalitz-plot studies. This consistency is important: changing the selection between notebooks would make comparisons ambiguous and could introduce artificial differences in the resulting distributions.

6 Signal and Background Classification

6.1 Sideband-labelled samples

The signal/background study uses mass windows to create approximate labels. Candidates in a narrow window around the B mass are labelled as signal-like, while candidates in lower and upper mass sidebands are labelled as background-like. The sideband method is common in HEP because sideband regions provide a data-driven representation of combinatorial background.

The label definition used in the notebook is

$$\text{signal}: \quad 5270 < m_B < 5290 \text{ MeV}/c^2, \tag{11}$$

$$\text{background}: \quad 5200 < m_B < 5250 \text{ MeV}/c^2 \quad \text{or} \quad 5310 < m_B < 5350 \text{ MeV}/c^2. \tag{12}$$

These labels are proxy labels. The signal window can still contain background, and the sidebands may not perfectly represent the background under the signal peak. Nevertheless, the construction is useful for demonstrating a realistic machine-learning workflow without relying on simulation truth information.

6.2 Random Forest classifier

A Random Forest classifier is trained using reconstruction-quality and PID variables. The B-candidate mass is deliberately excluded from the feature list, because the mass defines the labels. Including it would make the classifier learn the sideband construction rather than physically meaningful differences in candidate quality or PID response.

The classifier uses the following feature set:

$$\left\{\chi^2_{\text{vertex}}(B),\, d_{\text{flight}}(B),\, P_K(H_1), P_K(H_2), P_K(H_3),\, P_\pi(H_1), P_\pi(H_2), P_\pi(H_3)\right\}. \tag{13}$$

The trained model is evaluated on a held-out test sample. The receiver operating characteristic (ROC) curve in Fig. 4 shows the trade-off between signal efficiency and background misclassification rate. The diagonal line corresponds to a classifier with no discriminating power.

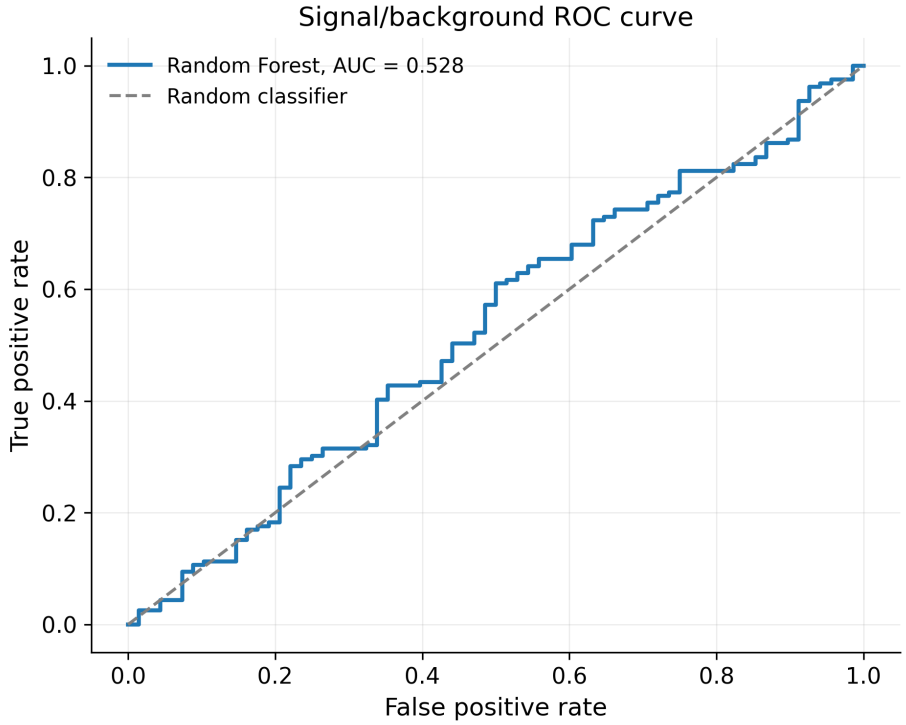


Figure 4: ROC curve for the sideband-labelled Random Forest signal/background classifier.

Figure 5 shows the feature-importance ranking from the Random Forest model. Such rankings are useful diagnostics, but they should not be overinterpreted as direct physics observables. They depend on feature correlations, sample composition, and the chosen model architecture.

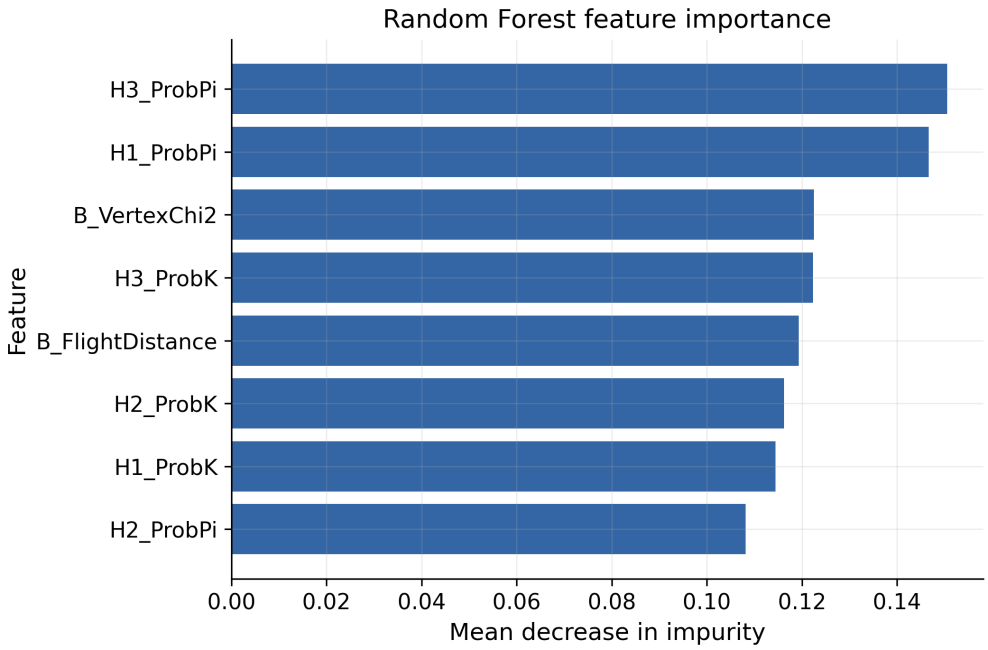


Figure 5: Random Forest feature importance for the signal/background classification task.

This part of the project demonstrates the integration of standard HEP sideband logic with modern machine-learning tools. Future improvements could include gradient-boosted decision trees, cross-validation, systematic studies of the sideband definition, and comparisons with simulation.

7 CP Asymmetry Study

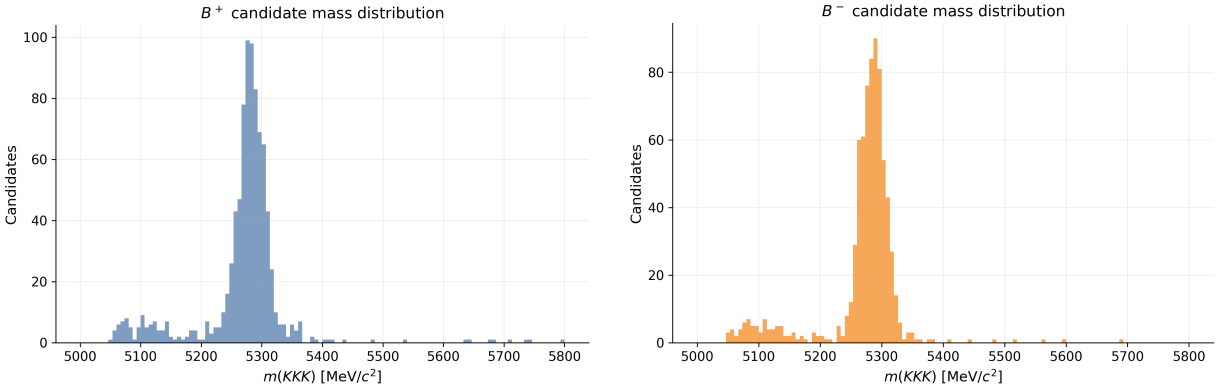
7.1 Charge separation

The CP-asymmetry notebook separates selected candidates according to the total charge of the three daughter tracks:

$$Q_{\mathrm{tot}} = Q(H_1) + Q(H_2) + Q(H_3). \tag{14}$$

Candidates with $Q_{\mathrm{tot}} = +1$ are classified as B^+ candidates, while candidates with $Q_{\mathrm{tot}} = -1$ are classified as B^- candidates. This charge assignment is natural for a three-charged-hadron final state and uses the track-charge information reconstructed by the tracking system.

Figure 6 shows the selected invariant-mass distributions for the two charge categories. Comparing these distributions checks whether the reconstruction and selection behave similarly for the two charge-conjugate samples.



(a) B^+ candidates

(b) B^- candidates

Figure 6: Selected charge-separated B-candidate invariant-mass distributions.

The overlaid comparison in Fig. 7 provides a compact visual summary of the charge-separated mass spectra. Similar peak shapes support the use of a common reconstruction and selection model, while differences in yield motivate the raw asymmetry calculation.

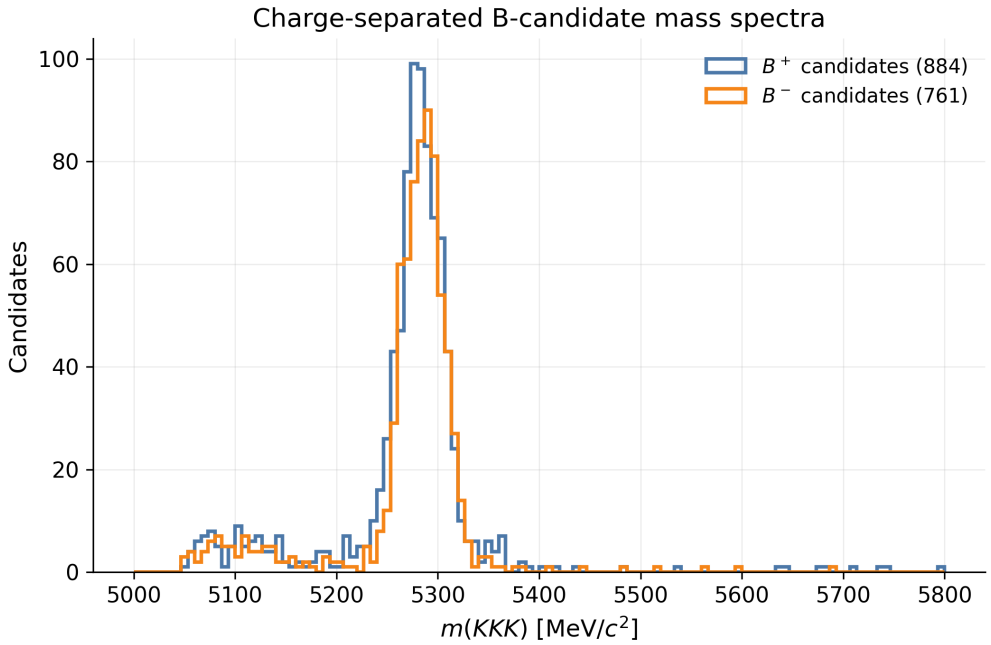


Figure 7: Overlay of selected B^+ and B^- candidate mass spectra.

7.2 Raw integrated asymmetry

The raw integrated asymmetry is computed in the signal window

$$5270 < m_B < 5290 \text{ MeV}/c^2. \tag{15}$$

Using the event counts obtained in the notebook,

$$N(B^+) = 282, \qquad N(B^-) = 246, \tag{16}$$

the raw asymmetry is

$$A_{\text{CP}}^{\text{raw}} = \frac{246 - 282}{246 + 282} = -0.06818. \tag{17}$$

This value should be interpreted carefully. It is not a corrected CP-asymmetry measurement. It still includes possible detector effects, production asymmetry, residual background, finite-sample fluctuations, and the use of a single magnet polarity. A realistic CP analysis would fit the B^+ and B^- mass spectra separately, subtract or model backgrounds, combine magnet polarities, and propagate systematic uncertainties. The current result is best viewed as a first diagnostic that demonstrates how a charge-asymmetry study is structured.

8 Dalitz Plot Analysis

8.1 Pairwise invariant masses

Three-body decays contain more kinematic information than a single reconstructed parent mass. The internal phase-space structure can be studied using pairwise invariant masses. For three daughter tracks labelled K_1, K_2, and K_3, the Dalitz variables used in this project are

$$m^2(K_1K_2), \qquad m^2(K_2K_3), \qquad m^2(K_1K_3). \tag{18}$$

Only two of these variables are independent for a fixed parent mass, but plotting different pairs is useful for visual inspection and for checking symmetries or structures involving different track orderings.

The pair mass is computed by summing the two daughter four-vectors:

$$m_{ij}^2 = (E_i + E_j)^2 - |\vec{p}_i + \vec{p}_j|^2 \,. \tag{19}$$

As in the B-mass reconstruction, each daughter is assigned the kaon mass hypothesis. The resulting squared masses are reported in GeV^2/c^4 to keep the numerical axis scales readable.

8.2 Dalitz distributions and resonance structure

A Dalitz plot displays the density of selected events in a two-dimensional plane of pairwise invariant-mass-squared variables. Resonant two-body substructures can appear as bands, while interference and phase-space effects can produce non-uniform densities. In a complete amplitude analysis, these structures would be modelled with coherent sums of resonant and non-resonant amplitudes. The present analysis restricts itself to visualising the distributions.

Figure 8 shows the distribution in the $m^2(K_1K_2)$ versus $m^2(K_2K_3)$ plane. The use of a two-dimensional histogram rather than a raw scatter plot makes dense regions easier to read.

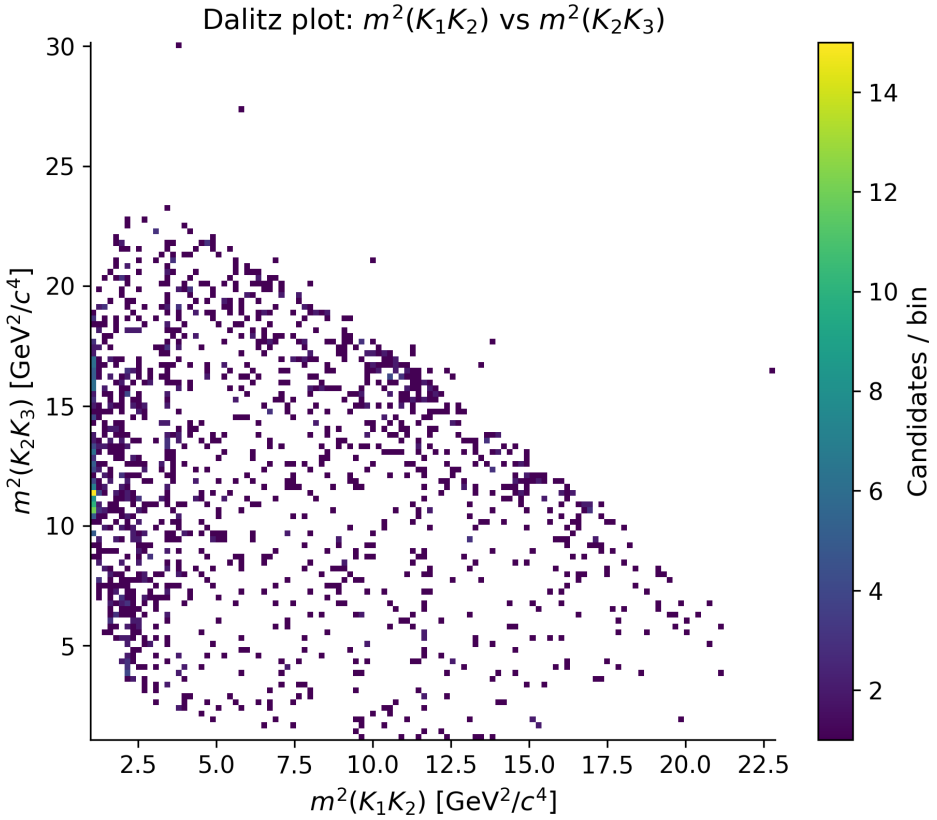


Figure 8: Dalitz density map in the $m^2(K_1K_2)$ versus $m^2(K_2K_3)$ plane.

Figures 9 and 10 show the complementary pairings. Together, these projections provide a first view of the three-body phase-space population after the baseline selection.

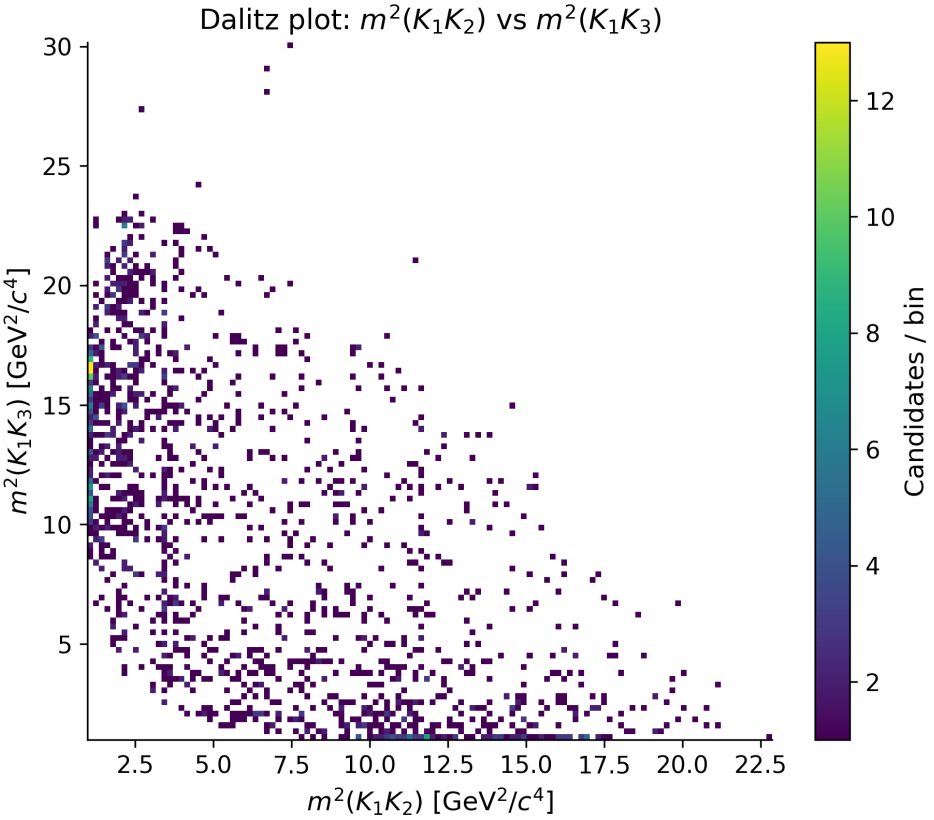


Figure 9: Dalitz density map in the $m^2(K_1K_2)$ versus $m^2(K_1K_3)$ plane.

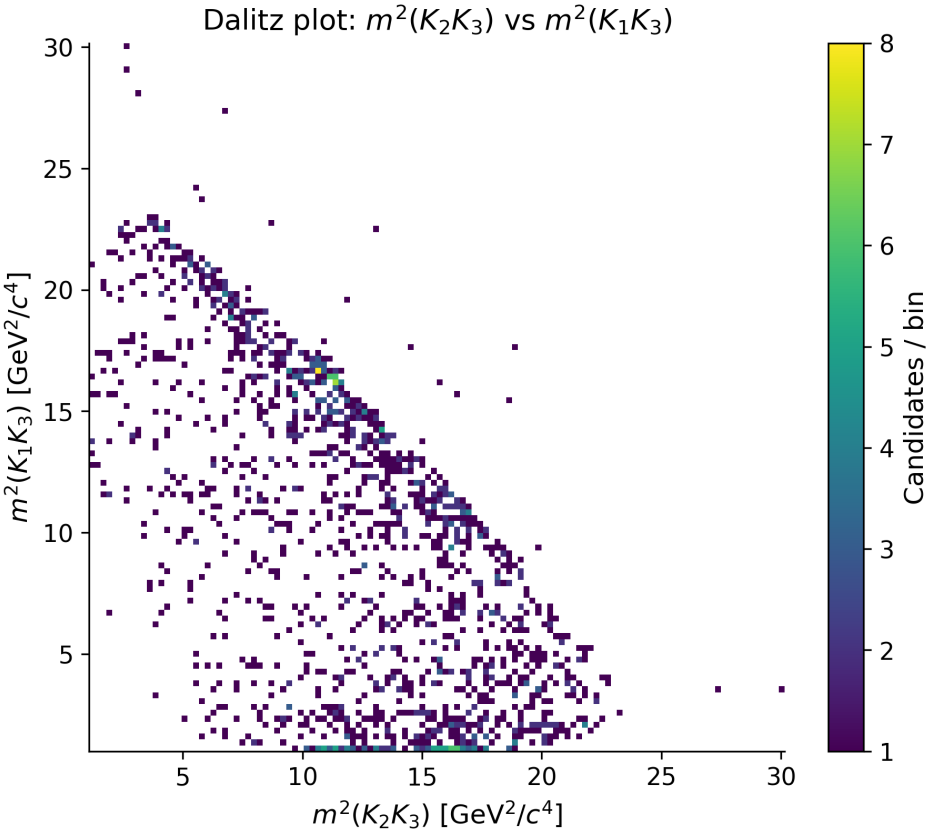


Figure 10: Dalitz density map in the $m^2(K_2K_3)$ versus $m^2(K_1K_3)$ plane.

Dalitz plots are especially relevant for CP-violation studies in multibody decays. Even if the integrated raw asymmetry is small, local asymmetries in restricted phase-space regions can be larger. A natural future extension would therefore compare the Dalitz distributions of B^+ and B^- candidates, using background subtraction and detector-efficiency corrections.

9 Conclusion

This project demonstrates a complete and modern introductory HEP analysis workflow using LHCb CERN Open Data. Starting from ROOT ntuples, the analysis reads selected branches with `uproot`, processes candidates with `pandas` and `numpy`, reconstructs invariant masses, applies physics-motivated selections, generates publication-style plots, trains a sideband-labelled Random Forest classifier, computes a raw charge asymmetry, and visualises three-body phase space with Dalitz plots.

The invariant-mass reconstruction shows a clear B-candidate peak after the baseline selection. A Gaussian-plus-linear fit provides a first signal-model validation. The signal/background classification study demonstrates how sideband labels can be combined with standard machine-learning tools, while the ROC curve and feature-importance plot provide useful performance diagnostics. The CP-asymmetry notebook separates B^+ and B^- candidates and computes a raw integrated asymmetry in the signal window. The Dalitz analysis extends the project toward multibody-decay phase-space studies by reconstructing pairwise invariant-mass-squared variables.

The analysis remains intentionally simplified. Several improvements would be required for a precision measurement:

- train stronger classifiers such as XGBoost or gradient-boosted decision trees, with cross-validation and robust overtraining checks;
- include detector effects, PID calibration, trigger efficiencies, and reconstruction asymmetries;
- use simulation and control samples to validate selection efficiencies and background shapes;
- combine MagnetUp and MagnetDown samples to reduce charge-dependent detector biases;
- fit B^+ and B^- mass spectra separately with more realistic signal and background models;
- perform local CP-asymmetry studies across Dalitz-plot regions.

Despite these limitations, the project captures the core logic of a real experimental-particle-physics analysis: transparent event reconstruction, documented selection choices, modular code, reproducible figures, and explicit discussion of systematic limitations. It therefore provides a strong foundation for more advanced studies in heavy-flavour physics and for future work in an experimental PhD environment.

References

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[3] Rene Brun and Fons Rademakers. ROOT: An object oriented data analysis framework. *Nuclear Instruments and Methods in Physics Research Section A*, 389(1–2):81–86, 1997.